14/7/2025 (AN)

Hall Ticket Number:

MR-21

Code: B315CB6S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July 2025

Computer Organization and Architecture

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What are register transfer logic languages? Explain RTL statements?	2	1	1	1
1. b)	Define machine instruction what are the elements of machine instruction?	2	1	1	1
1. c)	Draw the block diagram of a control memory and associated hardware needed for selecting the next microinstruction address.	2	6	2	1
1. d)	How to map micro operation to a micro instruction address.	2	1	2	2
1. e)	What are the various ways to represent the negative value in fixed point numbers?	2	1	5	1
1. f)	What is meant by normalization in floating point representation? Why do we need it?	2	1	5	1
1. g)	Compare synchronous and Asynchronous serial data transmission.	2	2	4	1
1. h)	Differentiate between Write through a cache policy and Write back a cache policy?	2	2	4	3
1. i)	Compare RISC & CISC design philosophy?	2	5	3	1
1. j)	Explain the static Inter process arbitration process?	2	2	3	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Demonstrate the functional units and components that are used for Understanding of computer Organization with neat sketch.	5	2	1	1
2. b)	Construct a circuit to addition, subtraction for 4 bit register using full adders and explain control logic?	5	3	1	3
	OR				
2. c)	Identify different types of instructions and describe their formats with their Constituent Fields?	5	3	1	1
2. d)	Design and implement a very simple CPU with respective RTL code and fetch decode cycles?	5	6	1	3
3. a)	What are the major design considerations in microinstruction sequencing?	5	1	2	2
3. b)	What are the design goals for a designer while deciding a hardwired or micro-programmed CU for a CPU?	5	1	2	1
	OR				
3. c)	Explain about microinstruction sequencing techniques, specifically variable format address microinstruction.	5	1	2	1
3. d)	What is addressing mode? List out the various addressing modes with suitable examples.	5	1	2	1
4. a)	Explain about sign magnitude and 2's complement approaches for representation fixed point numbers? Why 2's complement is preferable.	5	1	5	1
4. b)	What is overflow and underflow? If the computer is considered as infinite system does still have these problems?	5	5	5	2

P.T.O.

	OR				
4. c)	Perform 10111 * 10111 using 2's complement Multiplication algorithm.	10	3	5	2
5. a)	Design working principle of DMA controller with a neat diagram.	6	3	3	3
5. b)	Explain interrupt initiated I/O in detail.	4	1	3	1
	OR				
5. c)	Why do hierarchical memory systems are employed to achieve optimal performance by a computer system.	5	3	3	1
5. d)	What is a cache mapping function and explain various types of mapping techniques in cache memory organization.	5	2	3	1
6. a)	Explain Matrix multiplication in related with Vector Processing.	5	1	4	1
6. b)	List and explain various types of solutions for handling pipeline Hazards.	5	1	4	1
	OR				
6. c)	Design Time Shared common bus structure for multiprocessors with a neat sketch.	5	2	4	3
6. d)	Distinguish between tightly coupled microprocessors and tightly coupled Microprocessors.	5	4	4	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome